Wings, Lee Bounds and Tail Control

How four models keep their extrapolated wings arbitrage-admissible

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Abstract

A smile model earns its keep in the *wings* — the region beyond the quoted strikes, where the fit becomes pure extrapolation and where mispriced tails leak into var-swaps, risk-reversals and butterflies. This cross-cutting note collects the wing treatment shared by the Vol-Fitter's models. The universal constraint is Lee's moment formula: the asymptotic slope of total implied variance, $\beta = \limsup_{k\to\pm\infty} w(k)/|k|$, cannot exceed 2, and it is tied to the number of finite moments of the underlying by an explicit map $\beta = \psi(p)$. We show how each model realizes or enforces this: LQD reads the slopes off its endpoint scales and is Lee-consistent *by construction*; SVI has linear wings and enforces the bound with a soft penalty; Multi-Core SIV inherits its base wings because the hats are wing-neutral; and the local-vol surface extrapolates with a controlled linear slope. A single calibration window, fitted by all three parametric models and extrapolated far, shows their wings agreeing in sign and order of magnitude while differing in detail — and we explain the subtle bug in which a fixed wide arbitrage-floor grid let a model's own linear extrapolation manufacture phantom violations, and the confinement fix.

Contents

1 The universal constraint: Lee's moment formula

Let $w(k,T) = \sigma^2_{\mathrm{BS}}(k,T)\,T$ be total implied variance at log-moneyness $k = \log(K/F_T)$. Lee's theorem bounds how fast the wings may grow.

Theorem 1 (Lee moment formula)**.** *Define the right tail slope* $\beta_R = \limsup_{k\to+\infty} w(k,T)/k$ *and the left slope* $\beta_L = \limsup_{k\to-\infty} w(k,T)/|k|$*. Let* $p_R^* = \sup\{p : \mathbb{E}[S_T^{1+p}] < \infty\}$ *and* $q_L^* = \sup\{q : \mathbb{E}[S_T^{-q}] < \infty\}$ *be the right/left moment critical exponents. Then*

$$\beta_R = \psi(p_R^*), \qquad \beta_L = \psi(q_L^*), \qquad \psi(p) = 2 - 4\left(\sqrt{p^2+p} - p\right), \tag{1}$$

and in particular $0 \le \beta \le 2$*. Equivalently* $\frac{1}{2\beta} + \frac{\beta}{8} - \frac{1}{2} = p$*.*

Figure 1 plots ψ. It is decreasing, from $\psi(0) = 2$ (no finite moment beyond the first $\Rightarrow$ steepest admissible wing) to $\psi(\infty) = 0$ (all moments finite $\Rightarrow$ flat wing). The ceiling $\beta = 2$ is model-free: any smile with a steeper asymptotic variance slope admits arbitrage.

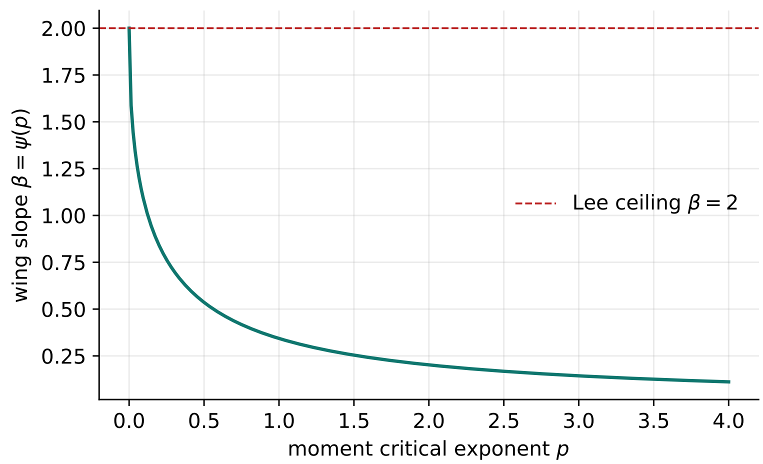

Figure 1: The Lee map $\beta = \psi(p)$: heavier tails (smaller moment exponent p) force steeper variance wings, capped at the model-free ceiling $\beta = 2$.

Heuristic

The wing slope and the tail heaviness are two faces of one coin. A fat right tail (few finite positive moments) *must* show up as a steep right variance wing; a model that draws a flat wing over a fat-tailed density is internally inconsistent and will misprice far-OTM calls. Lee's bound is therefore not a nuisance constraint but a consistency condition between the smile and the law it implies — which is why the LQD model, built from the law outward, satisfies it automatically.

2 Four models, four wing mechanisms

2.1 LQD: Lee-consistent by construction

The LQD model (Note 01) parametrizes the log quantile density with logarithmic endpoints, giving *exponential* return tails with scales A_L, A_R read off the endpoint coefficients. The moment exponents are then exact, $p_R^* = 1/A_R - 1$ and $q_L^* = 1/A_L$, and the wing slopes follow from theorem 1,

$$\beta_L = \psi(1/A_L), \qquad \beta_R = \psi(1/A_R - 1), \quad A_R < 1. \tag{2}$$

No penalty enforces this — it is structural. The only wing-related constraint is the hard $A_R < 1$ (a finite forward), handled by the soft barrier of Note 01. The benchmark fit yields $(\beta_L, \beta_R) = (0.0971, 0.0358)$.

2.2 SVI: linear wings with a soft cap

Raw SVI (Note 02) has exactly linear wings, $w(k) \sim b(1 \mp \rho)|k|$, so its slopes are $\beta_L = b(1 - \rho)$ and $\beta_R = b(1 + \rho)$ in closed form ($= (0.1093, 0.0364)$ here, though the figure's fitted slice differs slightly). Lee's bound is the explicit inequality $b(1 + |\rho|) \le 2$, enforced as one of the two soft no-arbitrage penalties (weight `sviPenaltyWeight`, cap `leeSlopeMax`$= 2$). On any admissible smile the penalty is zero; it only fences the optimizer out of the steep-wing region.

2.3 Multi-Core SIV: inherited wings

The Multi-Core SIV slice (Note 03) is a convex base plus R *zero-wing* hats. Because each hat and its first two derivatives vanish in both tails, the asymptotic wings depend only on the base, with z-space variance slopes $S_0 \mp 2K_0/\kappa_{P,C}$. Converting to total-variance k-space via $k = \sigma_{\text{ref}}\sqrt{t}\, z$, $w = tv$ gives

$$\beta_{L,R} = \frac{\sqrt{t}}{\sigma_{\text{ref}}} \left| S_0 \mp \frac{2K_0}{\kappa_{P,C}} \right|, \tag{3}$$

$= (0.1057, 0.0297)$ on the benchmark. Adding cores cannot change the wings — the entire reason for the zero-wing design.

2.4 Local volatility: controlled linear extrapolation

The piecewise-affine local-volatility surface (Note 04) has no closed-form implied-vol wing; beyond the deepest quoted strike $x_{\min}$ it extrapolates the *local* variance linearly with a slope tied to the first interior cell (parameter `leftWingSlopeMult`), rising toward $x \to 0$. The Dupire map then turns that into an implied-vol wing. The slope multiplier and, when a var-swap quote is present, a free left-extrapolation coefficient give the desk direct control of the put wing where there are no strikes to pin it.

2.5 Side by side

Figure 2 fits the same narrow calibration window with all three parametric models and extrapolates far past it. The wings agree in sign (the equity put skew: $\beta_L > \beta_R$) and order of magnitude, but diverge in detail — each model's tail is a property of *its* extrapolation law, not of the (identical) quotes. Table 1 collects the analytic slopes.

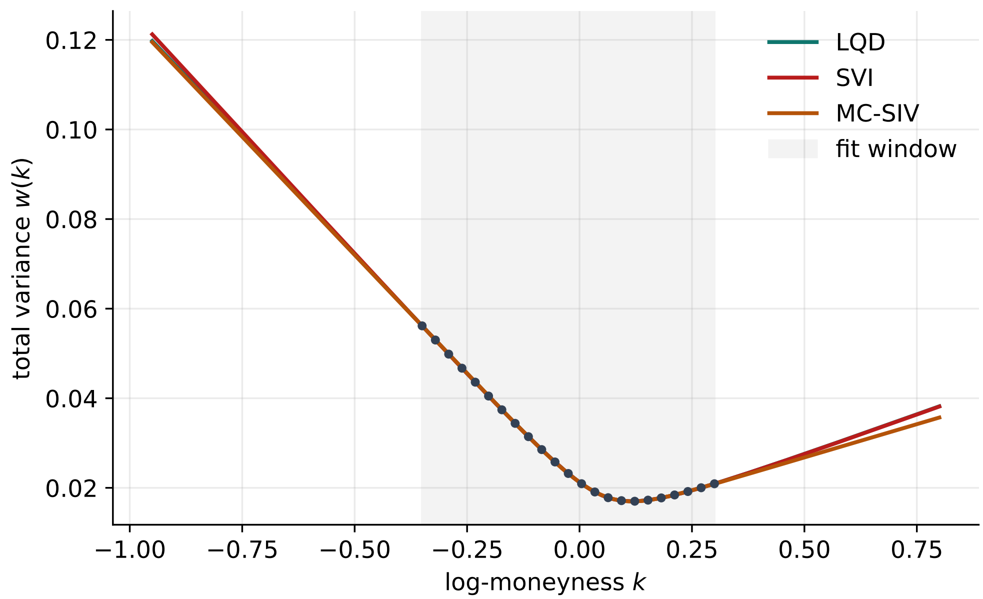

Figure 2: Three models fitted to the same window (shaded) and extrapolated. Inside the window they agree to vol bps; outside, each draws its own Lee-admissible wing.

Table 1: Analytic asymptotic total-variance wing slopes $\beta = \lim w(k)/|k|$ on the benchmark window; all are far below the Lee ceiling 2.

Model	β_L (put)	β_R (call)	mechanism
LQD	0.0971	0.0358	structural via $\psi(1/A)$
SVI	0.1093	0.0364	linear $b(1 \mp \rho)$, soft Lee cap
MC-SIV	0.1057	0.0297	base wings, hats neutral

3 A wing-extrapolation pitfall: phantom calendar violations

Wing extrapolation interacts dangerously with the *calendar* no-arbitrage machinery (Note 10). Calendar-freedom requires the later expiry's total variance to dominate the nearer one's at every k; the Vol-Fitter enforces this with a soft floor evaluated on a grid of log-moneyness points. The subtlety:

Caution

If the calendar floor is evaluated on a *fixed, wide* grid (say $k \in [-1, 1]$) that extends far beyond the traded strikes, then a model's own *linear-wing extrapolation* — which is unconstrained by data out there — can dip below the nearer expiry purely as an artefact of extrapolation, creating a *phantom* calendar violation. The optimizer then flattens the far fit to "cure" a problem the data never posed. This was observed live: SVI overlays on NVDA and SPY had their far wings flattened by phantom violations manufactured on the fixed wide grid.

The fix is *confinement*: the calendar floor is built only on the *traded* log-moneyness range (`variance_floor_grid_from` the actual quotes), so the constraint bites where there is data and the wings are left to each model's own Lee-admissible extrapolation. The LQD and sigmoid math were untouched; only the grid over which the floor is sampled changed. This is the single most important practical lesson about wings: *do not let a constraint sample the curve where only extrapolation lives.* Note 10 develops the calendar machinery in full.

4 What is original here

The Lee theory is classical; the contribution is the *uniform* treatment. The same wing slope β is computed three structurally different ways ((2), the SVI linear slopes, (3)) and shown to agree, so the choice of model is a choice of *how* to extrapolate, not *whether* the extrapolation is admissible. And the confinement fix of section 3 — recognizing that an arbitrage constraint must not be sampled in the extrapolation region — is a genuinely non-obvious engineering insight that the cross-model perspective makes visible.

A Hyperparameter atlas

Table 2: Wing-related knobs (cross-model).

Knob	Default	Role
`leeSlopeMax`	2.0	SVI Lee wing-slope cap $b(1+\lvert\rho\rvert) \le \beta_{\max}$.
`sviPenaltyWeight`	10^3	Weight of the SVI Lee/min-variance penalties.
`leftWingSlopeMult`	1.5	LV left local-variance extrapolation slope multiple (Note 04).
`lvVolCapMult`	3.0	LV local-vol cap as a multiple of max implied vol (Note 04).
`EPS_AR` (hidden)	10^{-6}	LQD $A_R < 1$ buffer; the only LQD wing constraint (Note 01).
calendar floor grid	traded range	Confined to quoted k to avoid phantom wing violations (Note 10).

Performance. Wing handling adds no measurable cost: LQD slopes are a by-product of the endpoint scales already computed; SVI/SIV slopes are closed forms of the fitted parameters; the LV wing is part of the surface solve. The only performance-relevant choice is the confinement of section 3, which *reduces* work by not over-constraining the extrapolation region.

References

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[3] J. Gatheral and A. Jacquier. Arbitrage-free SVI volatility surfaces. *Quantitative Finance*, 14(1):59–71, 2014.